

GROUP no



LECTURE CHALLENGE – write the code in PYTHON for computing *PMV*, *PPD* and *t_o* for the considered day in the previous simulation (exterior air parameters).

Save it as `calculate_pmv_ppd(t_a, t_mr, v=, rh=, i_clo, m_met).py`

Results should appear in the form: 'PMV=', 'PPD=', 't_o=', 'n iterations='.

Then, find interior comfort conditions so that $-0.5 < \text{PMV} < 0.5$ (change t_a , t_{mr} , RH).

Summer day in Romania (exterior air)	Interior air conditions
$t_a = 30^\circ\text{C}$	$t_a = \text{ }^\circ\text{C}$
$t_{mr} = 30.5^\circ\text{C}$	$t_{mr} = \text{ }^\circ\text{C}$
$RH=\varphi = 60\%$	$RH=\varphi = 60\%$
$v = 0.15 \text{ m/s}$	$v = 0.15 \text{ m/s}$
$\text{clo} = 0.5$	$\text{clo} = 0.5$
$M_{\text{met}} = 1.2 \text{ met}$	$M_{\text{met}} = 1.2 \text{ met}$
PMV = 1.767	PMV =
PPD = 65.3	PPD =
$t_o = 30.25 \text{ }^\circ\text{C}$	$t_o = \text{ }^\circ\text{C}$
n iterations = 8	n iterations =

Algorithm for Calculating PMV and PPD Indices (Based on ISO 7730)**Step 1: Initialization of Environmental and Personal Variables**

The algorithm requires six **primary input parameters** collected from sensors and occupant profiles:

- t_a : air temperature (°C)
- t_{mr} : mean radiant temperature (°C)
- v : air velocity (m/s)
- RH : relative humidity (%)
- I_{clo} : clothing insulation (clo)
- M_{met} : metabolic rate (met)

Step 2: Unit Conversions and Preliminary Calculations

The standard comfort models require standard **metric units** (W/m^2 for energy and m^2K/W for insulation).

- **Metabolic rate conversion:** $M = M_{met} \cdot 58.15$
- **Clothing insulation conversion:** $I_{cl} = 0.155 \cdot I_{clo}$
- **Partial water vapor pressure (p_v):** Calculated using the Antoine equation for saturated vapor pressure.

$$p_{sat}(t_a) = e^{16.6536 - \frac{4030.183}{t_a + 235}} [kPa]$$

$$p_v [Pa] = \frac{RH[\%]}{100} \cdot (p_{sat}(t_a) \cdot 1000) [Pa]$$

- **Clothing surface area factor (f_{cl})** = the ratio of the surface area of the clothed body to the surface area of the nude body.

$$f_{cl} = \begin{cases} 1 + 1.29 \cdot I_{cl}, & I_{cl} < 0.078 \\ 1.05 + 0.645 \cdot I_{cl}, & I_{cl} \geq 0.078 \end{cases}$$

Step 3: Iterative Calculation of Clothing Surface Temperature (t_{cl})

The temperature of the clothing surface cannot be solved directly; it is determined through a numerical iteration (successive approximation) balancing the heat conducted through the clothing with the heat lost to the environment via radiation and convection.

1. **Initial estimation:** A starting absolute temperature T_{cl} is approximated based on internal variables.
2. **Convective heat transfer coefficient (h_c):** The algorithm evaluates both forced convection ($h_{c,f}$) and natural convection ($h_{c,n}$) and selects the dominant heat transfer mode at each iteration:
 - o Forced convection: $h_{c,f} = 12.1 \cdot v^{0.5}$
 - o Natural convection: $h_{c,n} = 2.38 \cdot |t_{cl} - t_a|^{0.25}$
 - o Selection: $h_c = \max(h_{c,f}, h_{c,n})$
3. **Iteration:** The value of t_{cl} is recalculated until the difference between successive iterations is less than a threshold error ($\epsilon = 0.0015$).

Step 4: Human Body Heat Loss Components

Once t_{cl} and h_c converge, the algorithm calculates the six distinct paths of heat loss from the human body to the environment:

1. **Skin diffusion loss (HL_1):** Moisture diffusing through the skin

2. **Sweat loss** (HL_2): Evaporative heat loss from regulatory sweating (only if $M > 58.15 \text{ W/m}^2$)

$$HL_1 = 3.05 \cdot 10^{-3} \cdot (5733 - 6.99 \cdot M - p_v)$$

$$HL_2 = 0.42 \cdot (M - 58.15)$$
3. **Latent respiration loss** (HL_3): Heat lost through moisture exhaled from the lungs

$$HL_3 = 1.7 \cdot 10^{-5} \cdot M \cdot (5867 - p_v)$$
4. **Dry respiration loss** (HL_4): Sensible heat lost heating the inhaled air

$$HL_4 = 0.0014 \cdot M \cdot (34 - t_a)$$
5. **Radiation loss** (HL_5): Heat radiated from the clothing surface to the surrounding room surfaces (temperatures are in Kelvin: T_{cl} and T_{mr})

$$HL_5 = 3.96 \cdot f_{cl} \cdot \left[\left(\frac{T_{cl}}{100} \right)^4 - \left(\frac{T_{mr}}{100} \right)^4 \right]$$
6. **Convection loss** (HL_6): Heat convected from the clothing surface to the room air

$$HL_6 = f_{cl} \cdot h_c \cdot (t_{cl} - t_a)$$

Step 5: Calculation of **PMV and PPD**

The thermal sensation transfer coefficient (TS) correlates the internal metabolic heat production to the skin's thermal sensitivity

$$TS = 0.303 \cdot e^{-0.036 \cdot M} + 0.028$$

Net thermal load on the body (internal heat production minus all heat losses):

$$L = M - HL_1 - HL_2 - HL_3 - HL_4 - HL_5 - HL_6$$

The **Predicted Mean Vote (PMV)** is calculated by: $PMV = TS \cdot L$

The **Predicted Percentage Dissatisfied (PPD)** is derived statistically from the PMV index:

$$PPD = 100 - 95 \cdot e^{-0.03353 \cdot PMV^4 - 0.2179 \cdot PMV^2}$$

Step 6: Calculation of Operative Temperature (t_o)

$$t_o = \begin{cases} 0.5 \cdot t_a + 0.5 \cdot t_{mr}, & v < 0.2 \text{ m/s} \\ 0.6 \cdot t_a + 0.4 \cdot t_{mr}, & 0.2 \frac{m}{s} \leq v < 0.6 \text{ m/s} \\ 0.7 \cdot t_a + 0.3 \cdot t_{mr}, & v \geq 0.6 \text{ m/s} \end{cases}$$